

Attitudes and Behavior

Outline

1) What are attitudes?
    - Components of attitudes
    - How do we measure attitudes?
2) The attitude-behavior link
    - Attitudes to behavior
    - Behavior to attitudes
3) Cognitive dissonance theory
4) Self-perception theory

(1) Attitudes

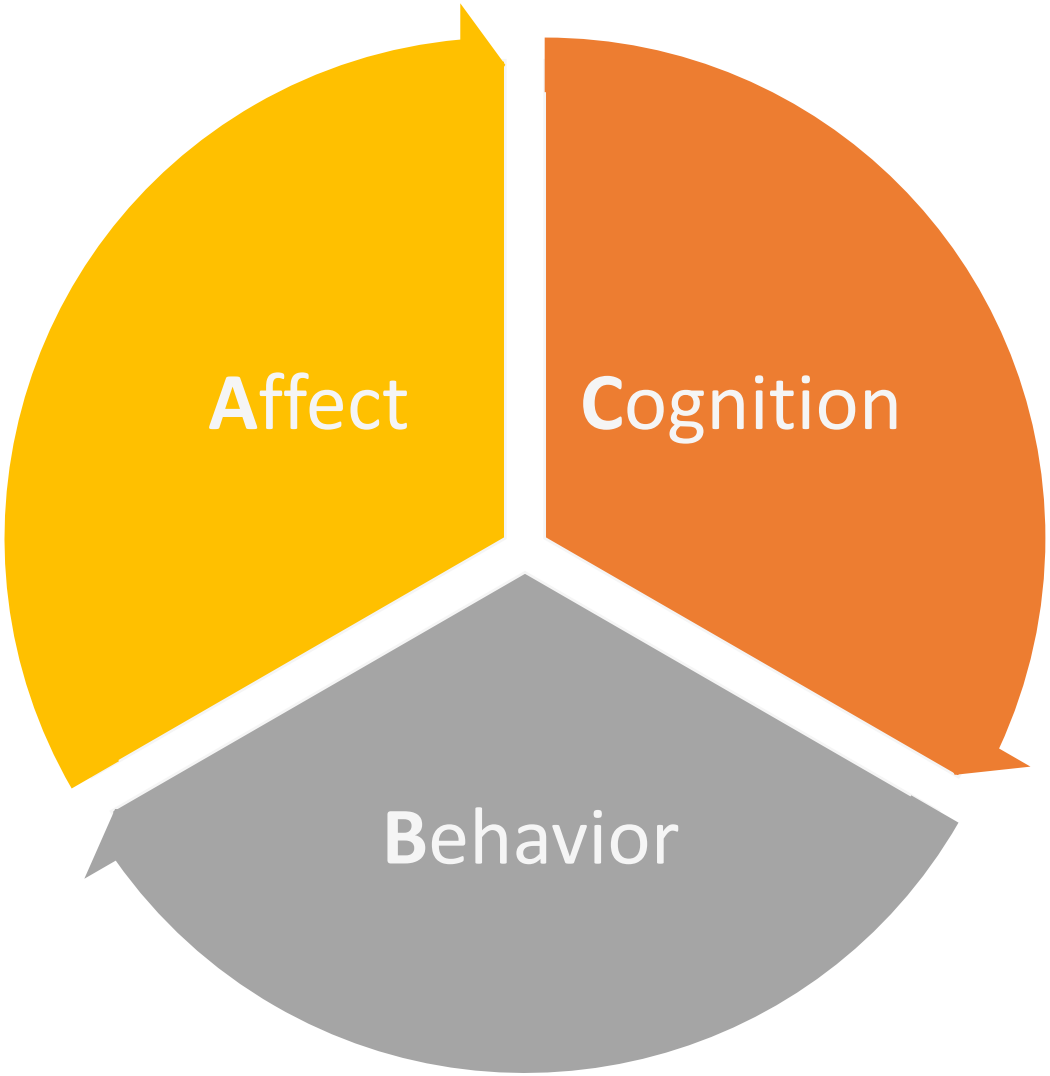
Attitudes



Attitudes

An attitude is an evaluation of an object along a positive-negative dimension.

Components of Attitudes



Components of Attitudes

- Affect/Emotion (A)

  Nearly every object triggers some degree of positive or negative emotion, which constitutes the affective component of the attitude somebody has toward it.

Components of Attitudes



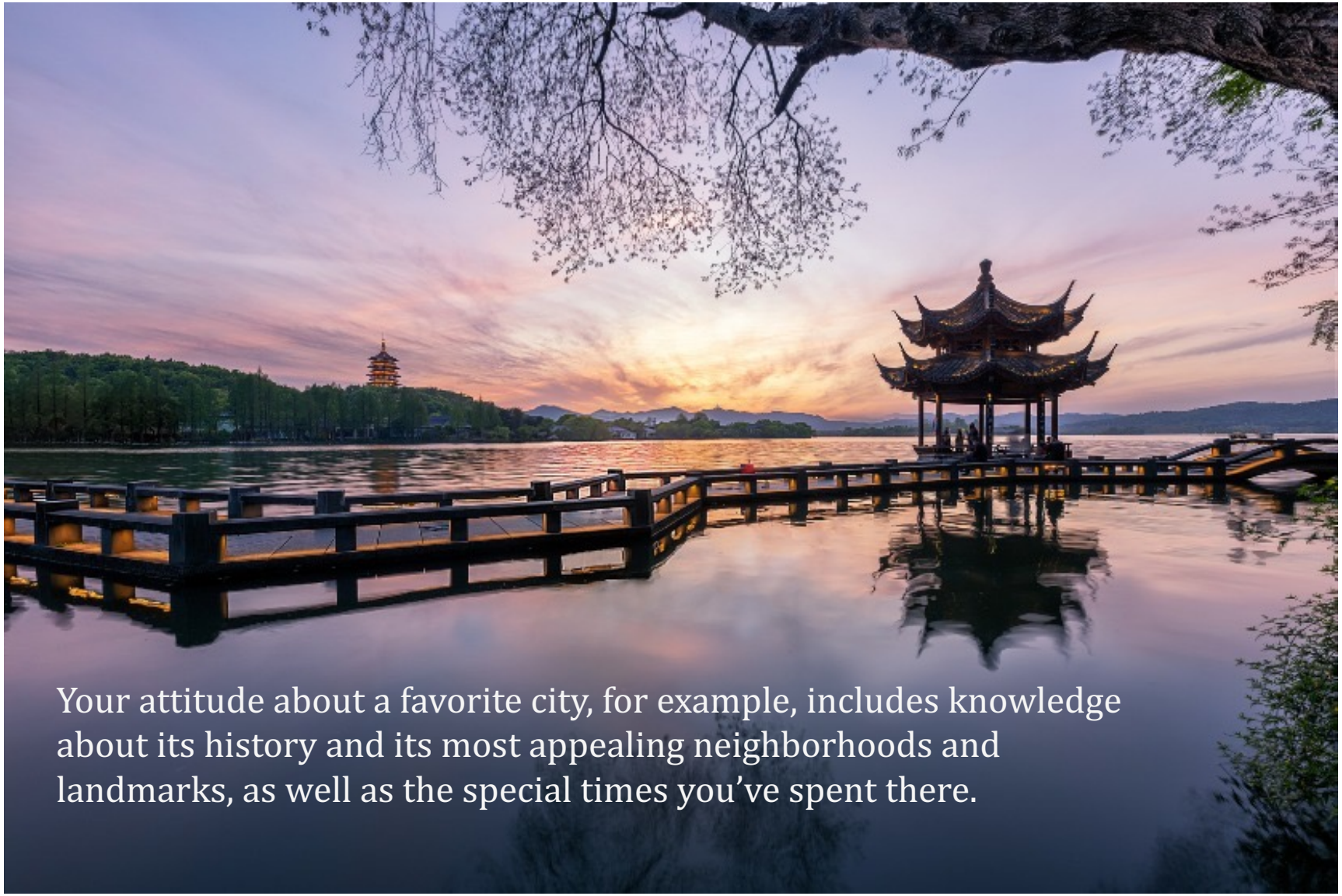


Components of Attitudes

- Cognitions (C)

  Includes knowledge and beliefs about the object, as well as associated memories and images.

Components of Attitudes



Your attitude about a favorite city, for example, includes knowledge about its history and its most appealing neighborhoods and landmarks, as well as the special times you've spent there.

Components of Attitudes

- Behaviors (B)

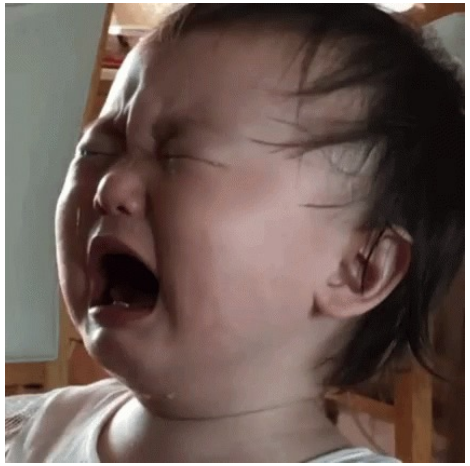
  The affective evaluation of good versus bad is connected to a behavioral tendency to either approach or avoid.

  - When specific attitudes are primed, even unconsciously——people are more likely to behave in ways consistent with the attitude (Chen & Bargh, 1999)

Components of Attitudes

Neuroscientific studies indicate that our attitudes activate particular brain regions, areas of the motor cortex, that support spcific actions.









Components of Attitudes

ABCs of attitudes

(affect, behavior, cognitions)

How do we measure attitudes?



How do we measure attitudes?

Example of Explicit Measures

- **A Likert Scale**: A numerical scale used to assess people's attitudes; a scale that includes a set of possible answers with labeled anchors on each extreme.

Attitudes toward the use of cell phones while driving

It's never acceptable — It's always acceptable

1 2 3 4 5 6 7

Likert Scale

How much do you value freedom?

How important is a less polluted environment?

How strongly do you feel about the need to reduce discrimination?



Accessibility - Response Latency

The accessibility of the attitude ––– how readily it comes to mind

Measure:

Response Latency: *The time it takes a person to respond to an attitude question*

- Someone who takes less than a second to respond is likely to have a stronger attitude than someone who takes several seconds to respond.

Centrality

The centrality of the attitude to the person's belief system

Measure:

A variety of attitudes within a domain and calculate how strongly each one is linked to the others

- For example, suppose you highly value environmental protection. You may find that you also hold strong attitudes on other related issues, such as opposing the use of plastic bags, supporting green energy policies, and boycotting polluting companies. Here, environmental protection is a high-centrality attitude in your belief system, influencing your views on many other related issues.

How do we measure attitudes?

在课堂上使用手机是否可接受？

How do we measure attitudes?

Explicit vs. Implicit Measures

How do we measure attitudes?

- Why is this distinction important?
  - Many of the attitudes that we study are socially sensitive such as attitudes toward social categories, religion, sex, and so people may be wary of giving their attitudes or they may sometimes not even know their nonconscious attitudes.
  - With many attitudes there is a big difference between how people respond on an implicit measure and an explicit measure.

How do we measure attitudes?

Unwilling or **Unable** to report true feelings or opinions

Is it possible that we are not even aware of our implicit attitudes?

当问一个每天要抽四包烟的吸烟者“要抽多少烟？”时，他可能报告每天只抽两包烟，因为承认抽四包烟可能会让自己有窘境。或者，吸烟者不会简单地回答这个问题，而是借口这是私人问题。这表明人们有时知道自己的看法但 **不愿意** 报告。

但也有可能那个吸烟者报告只抽两包烟，因为他在意识中认为自己只抽两包烟。这种无意识地给出错误答案的行为有时候被称为自我欺骗。这是因为他们 **没能力** 给出理想的答案，而采用自我欺骗的防御机制。

这种 **不愿意-没能力** 的区别就像有意隐藏什么不让别人知道和无意识中隐藏了什么连自己都不知道。

How do we measure attitudes?

Implicit Measures (1 min)

How do we measure attitudes? - Implicit Measures

- **Physiological Measures**
  - EEG and fMRI (brain activity)
  - ECG (heart rate)
  - Eye tracker (eye movement)
- **Nonverbal Behavior Measures**
  - Facial Expressions
  - Gestures
- **Reaction Time Measures**
  - Implicit Association Test (IAT)



IAT

https://implicit.harvard.edu/implicit/

BLACK/WHITE IAT

BLACKS

WHITES



BLACK/WHITE IAT

BLACKS

WHITES



BLACK/WHITE IAT

POSITIVE

NEGATIVE

love

BLACK/WHITE IAT

POSITIVE

NEGATIVE

War

BLACK/WHITE IAT

BLACKS
OR
POSITIVE

WHITES
OR
NEGATIVE

love

BLACK/WHITE IAT

BLACKS
OR
POSITIVE

WHITES
OR
NEGATIVE



BLACK/WHITE IAT

WHITES

BLACKS



BLACK/WHITE IAT

WHITES

BLACKS



BLACK/WHITE IAT

WHITES
OR
POSITIVE

BLACKS
OR
NEGATIVE

love

BLACK/WHITE IAT

WHITES
OR
POSITIVE

BLACKS
OR
NEGATIVE



Calculating IAT Scores

Pleasant & Black - one key / Unpleasant & White – other key

Minus

Unpleasant & Black – one key / Pleasant & White – other key

Hypothetical Examples of Ways to Classify IAT Scores

- 500 ms – 350 ms = 150 ms (strong preference for Whites)
- 450 ms – 350 ms = 100 ms (moderate preference for Whites)
- 400 ms – 350 ms = 50 ms (slight preference for Whites)
- 350 ms – 350 ms = 0 (no preference)
- 350 ms – 400 ms = -50 ms (slight preference for Blacks)
- 350 ms – 450 ms = -100 ms (moderate preference for Blacks)
- 350 ms – 500 ms = -150 ms (strong preference for Blacks)

Black and White IAT Scores

Your results suggest:

- 500 ms – 350 ms = 150 ms (strong preference for Whites)
- 450 ms – 350 ms = 100 ms (moderate preference for Whites)
- 400 ms – 350 ms = 50 ms (slight preference for Whites)

- 350 ms – 350 ms = 0 (no preference)

- 350 ms – 400 ms = -50 ms (slight preference for Blacks)
- 350 ms – 450 ms = -100 ms (moderate preference for Blacks)
- 350 ms – 500 ms = -150 ms (strong preference for Blacks)

Overview of IAT Procedure

1. 呈现概念词
2. 呈现属性词
3. 联合任务一
4. 对联合任务一进行测试
5. 让被试对概念词做出相反的判断，交换左右键反映的内容
6. 联合任务二
7. 对联合任务二进行测试

Example: Explicit vs. Implicit Measures

Greenwald, McGhee, & Schwartz (1998)

- Participants:
  - 26 White participants
- Measures:
  - Implicit Measure of Bias – IAT
  - Explicit Measure of Bias - Semantic Differential Scale

Blacks (Whites)

Bad____ : ____ : ____ : ____ : ____Good
Worthless ____ : ____ : ____ : ____ : ____Valuable
Unpleasant____ : ____ : ____ : ____ : ____ Pleasant

Example: Explicit vs. Implicit Measures

Implicit measure - IAT
25 of 26 people showed a preference for Whites – 96%

Explicit measure - Semantic Differential Scale
5 of 26 people showed a preference for Whites – 27%

Correlation of IAT with 5 explicit measures, r = .14

Implicit Association Test (IAT)



Are the IAT scores valid?

- Familiarity
- Perceptual salience asymmetries
- Mere cultural knowledge irrespective of personal endorsement of that knowledge
- …

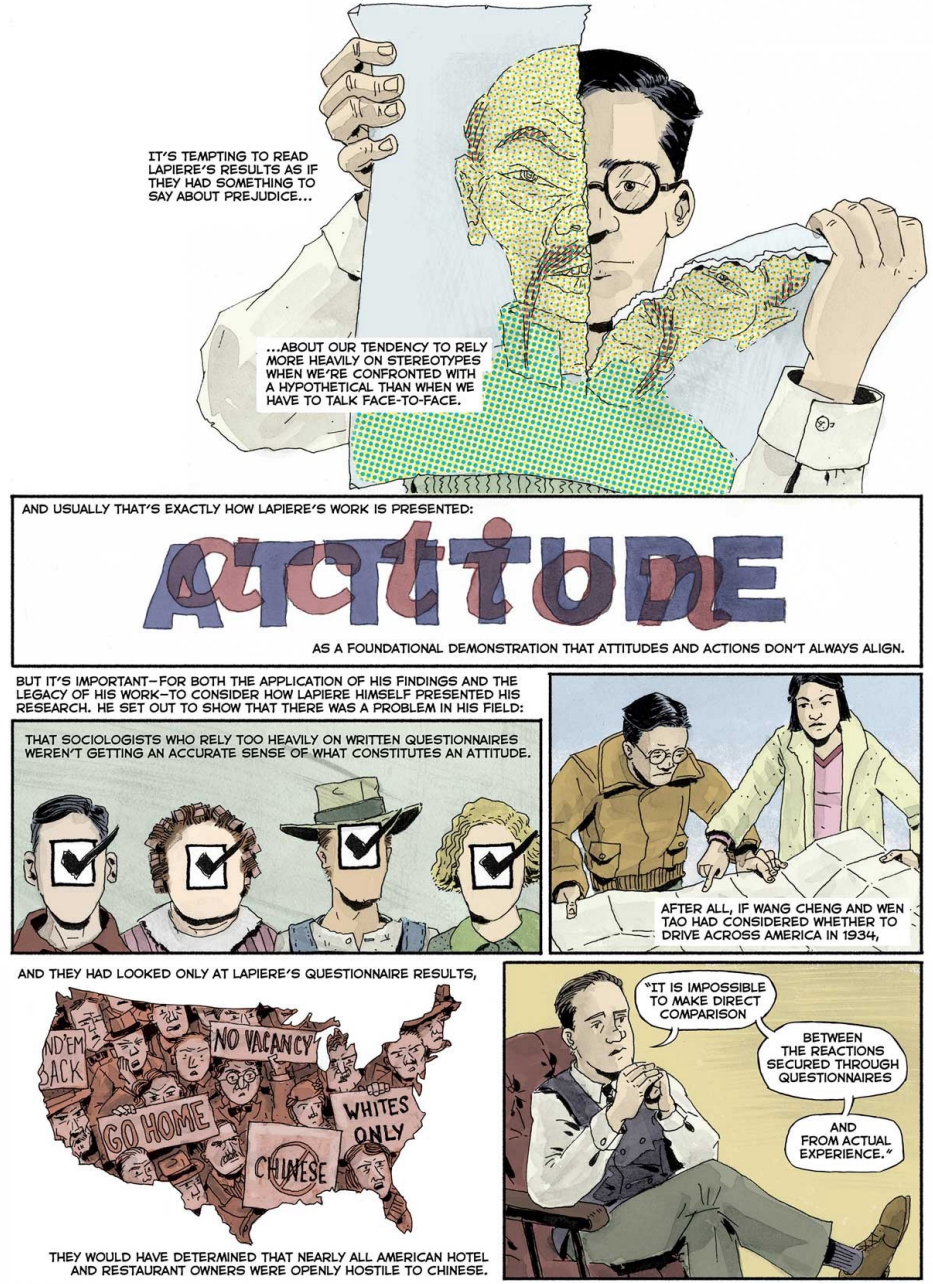
According to an article in The New York Times,
"there isn't even that much consistency in the same person's scores if the test is taken again".

(2) The Attitude-Behavior Link

The Attitude-Behavior Link

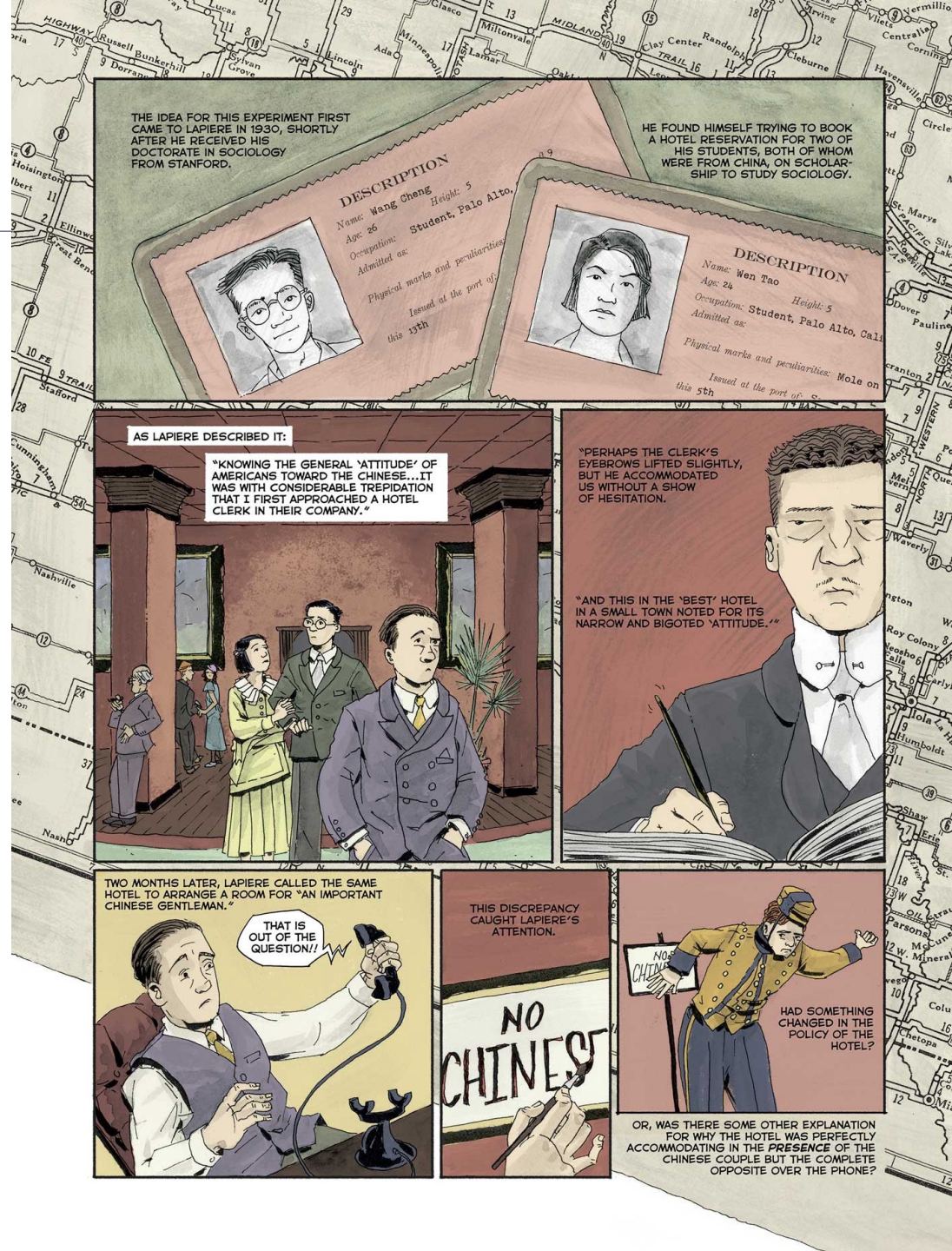
Do attitudes determine behaviour?

In **1931**, a Stanford sociologist toured the country with two Chinese students to test a hypothesis: **Do attitudes about racial discrimination differ from actions?**



The Attitude-Behavior Link

- **Study:**
  - **Real behavior**: from 1930 to 1933, they visited 67 hotels and 184 restaurants.
  - **Symbolic behavior:** Lapierre sent his letters to those establishments visited by travelers. “Do you agree to accept representatives of Chinese nationality as guests in your establishment?”
- Results:
  - **Questionnaires: 81** restaurants and **47** different hotels completed questionnaires. **NO** answered in **75** restaurants and **43** hotels.
  - **In practice:** there was only **one** refusal.

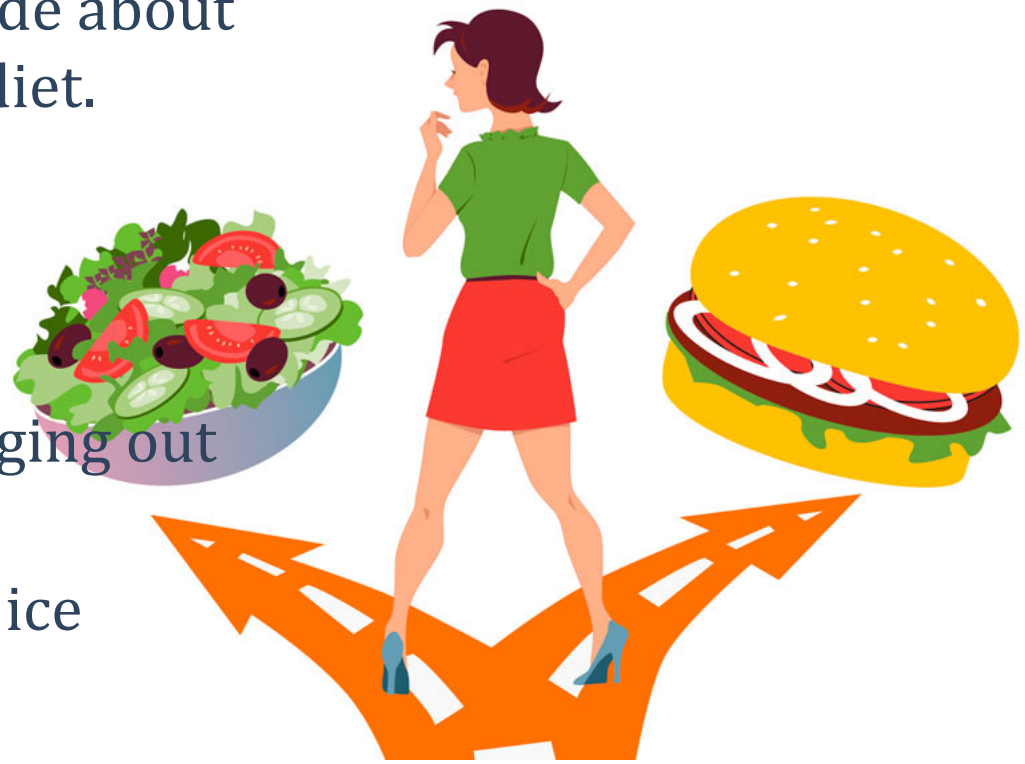


The Attitude-Behavior Link

Why do attitudes sometimes not predict behavior?

Reasons why attitudes sometimes do not predict behavior

The relationship between a person's attitude about dieting and actual success in sticking to a diet. Would you expect a strong relationship?



- Eating habits
- Whether a roommate happens to be pigging out at the moment
- The person's attitudes about things like ice cream, doughnuts, and French fries
- …

Reasons why attitudes sometimes do not predict behavior

1) There may be conflicting attitudes and other powerful determinants of behavior and other conflicting attitudes.
    - Conflicting attitudes towards other issues
    - Other powerful determinants
        - Habits
        - Situation
    - Perceived norms of appropriate behavior

Reasons why attitudes sometimes do not predict behavior

You might be dying to share your hilarious commentary about the movie with the guy next to you in the theater, but let's hope you refrain from doing so because you recognize it just isn't done and others would disapprove.



Reasons why attitudes sometimes do not predict behavior

2) Mismatch between attitudes and behaviours in automaticity level

Example - Dual Attitudes (Gut Feelings vs. Analysis)

- Why are you attracted to that person?
  - She is cute.
  - She is ambitious.
  - He is fun to be with.
  - …



Example - Gut Feelings vs. Analysis

Wilson and colleagues (1984)

- How happy are you in your relationship?
  - In 1st group: Participants gave an overall evaluation of their relationship.
  - In 2nd group: Participants first listed the reasons they felt the way they did and then gave an overall relationship satisfaction.
- Participants were contacted 9 months later.



Example - Gut Feelings vs. Analysis

- The attitudes of participants in the 1st group were much more accurate predictors of their current relationship status than the attitudes of participants in the 2nd group.

Why?

Example - Gut Feelings vs. Analysis

- By dissecting the relationship and analyzing it, people drew attention to easily verbalized cues and then based their attitudes on these cues. These reasonable cues, however, may not have been that influential in affecting our behaviors or even related to our attitudes.
- Our gut feelings or automatic attitudes may at times be more predictive of our true feelings and better predict behavior.

Example - Gut Feelings vs. Analysis

- Does this mean that introspection is always harmful? Should we always just go with our gut?

Not at all!

Example - Gut Feelings vs. Analysis





Reasons why attitudes sometimes do not predict behavior

3) Mismatch between attitudes and behaviours in specificity level
    - General attitudes predict behaviours in general
    - Specific attitudes predict specific behaviours

Example:

A person's religious attitude may not predict whether they go to church next weekend but it will predict the total number of a wide array of religious behaviours over time.

General attitudes predict behaviors in general

A general attitudes toward an object or a concept may not predict any specific behavior but if we average behaviours over many occasions, attitudes can predict behaviour.

Reasons why attitudes sometimes do not predict behavior

4) Attitudes differ in strength - not all attitudes are equal
    - Strong attitudes are more accessible. More accessible attitudes direct behavior.

Summary of reasons why attitudes sometimes do not predict behavior

1) There are other powerful determinants of behavior and other conflicting attitudes.
2) Mismatch between attitudes and behaviours in automaticity level.
3) Mismatch between attitudes and behaviours in specificity level.
4) Attitudes differ in strength.

Do attitudes determine behavior?

Sometimes.

Does behavior determine attitudes?

Sometimes.

Why and when do behaviors change attitudes?



Example:

- Many young people resent being sent to church, temple, mosque, or other religious service and often complain, “Why do I have to go? I don’t believe any of this stuff.”
- Many of them are told, “It doesn’t matter if you believe it. What’s important is that you continue with your studies and your prayers.”

- Some resist to the very end and abandon all religious rituals and practices the minute their parents give them permission to opt out.
- But **a remarkable number stick with it** and eventually find themselves genuinely holding some of the very religious convictions they once resisted.

Why and when do behaviors change attitudes?

- Cognitive Dissonance Theory
- Self-Perception

(3) Cognitive Dissonance Theory

Cognitive Dissonance Theory

- The theory that inconsistency between a person's thoughts, sentiments, and actions creates an aversive emotional state (dissonance) that leads to efforts to restore consistency.

Leon Festinger, 1957

Cognitive Dissonance

- For example, in pursuit of a more comfortable living experience, you decide to move from a mini apartment in the city center to a larger house in the suburbs. However, this change results in a longer commute time of one hour by car, compared to the previous ten-minute walk to work.

- According to Festinger, once you've made an irrevocable decision, you'll exert effort to reduce this dissonance.

    - You'll rationalize.
    - You'll tell yourself and your friends how much you've learned from the audiobooks you plays in your car during your long commute.
    - …

Cognitive Dissonance

- Knox and Inkster (1968) interviewed bettors at a racetrack, some just before and some just after placing their bets.
    - Bettors who were interviewed right before they placed their bets gave their horses, on average, a “fair” chance of wining.
    - Bettors who were interviewed right after placing their bets gave their horses, on average, a “good” chance to win.



Cognitive Dissonance



- Frenkel and Doob (1976)

  Voters express greater confidence in their candidates when interviewed after they have voted than when interviewed right beforehand.

When does the dissonance occur?

- Festinger argued that dissonance reduction takes place only **after an irrevocable decision** has been made.

- Later research suggests that the same sorts of rationalization and distortion that occur after people make a decision also subconsciously take place **before** they make the decision.

- Once people develop a slight preference for one option over the others, they distort subsequent information to support their preference (Brownstein, 2003).

Cognitive Dissonance

Under what circumstances do you often experience cognitive dissonance?

You pay a high price for something (money, time, or effort), and it turns out to be disappointing.

Effort Justification

- Cognitive Dissonance → **Effort Justification**: The tendency of devoting mental energy to justifying what you've done.

Effort Justification





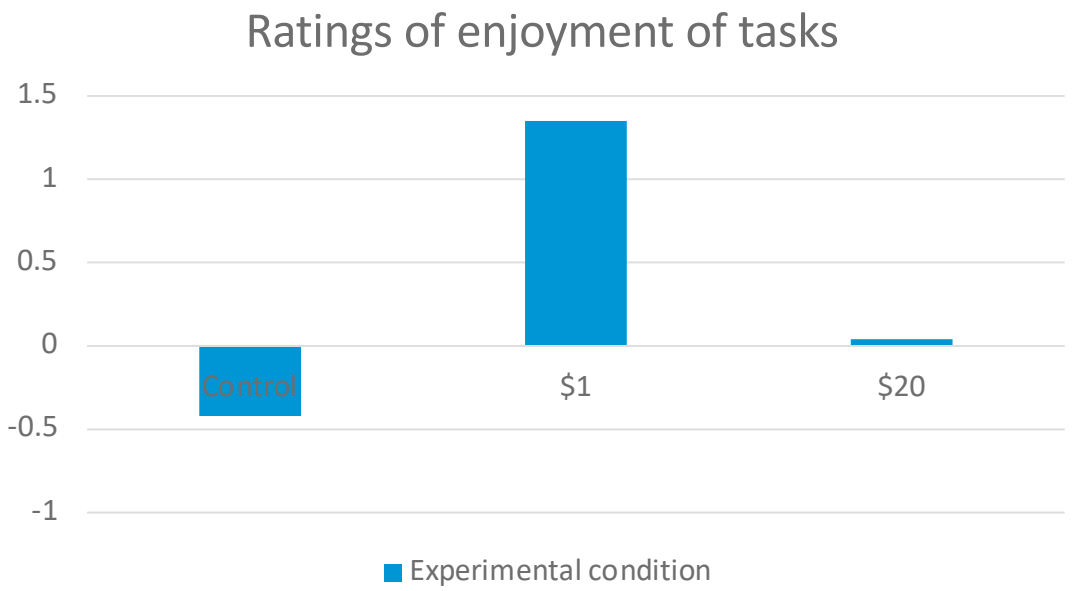
Cognitive Dissonance

Under what circumstances do you experience cognitive dissonance?

- When people are induced to behave in a manner that is inconsistent with their beliefs, attitudes, or values.

Induced Compliance and Attitude Change

- 3 conditions:
  - **Control**: completing a boring task and rating enjoyment
  - **$1**: paid 1$ to telling the next participant this study was very interesting.
  - **$20**: paid 20$ to telling the next participant this study was very interesting.



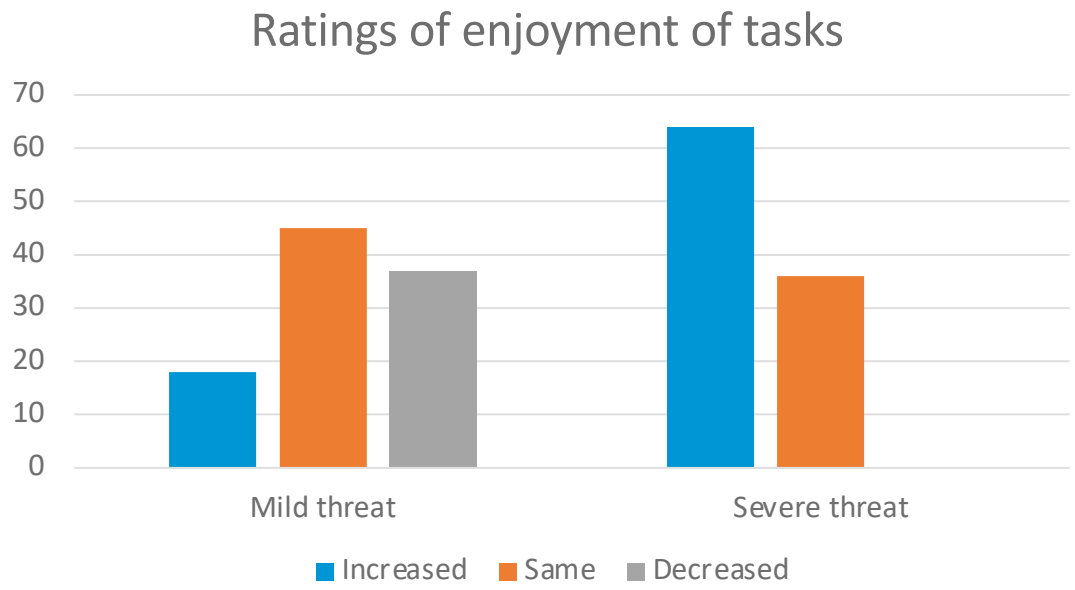
(Festinger & Carlsmith, 1959)

Induced Compliance and Undesired Behavior

"Forbidden toy" paradigm

- Free to play with any of the 5 toys except his/her second-favorite.
  - Mild threat: annoyed
  - Severe threat: very angry and would have to take all the toys and go home and never come back again.





Induced Compliance

Subtly compelling people to behave in a manner that is inconsistent with their beliefs, attitudes, or values in order to elicit dissonance and therefore a change in their original attitudes and values.

Summary of Cognitive Dissonance Theory

- Cognitive Dissonance Theory

❖ Effort Justification

❖ Induced Compliance

(4) Self-Perception Theory

Self-Perception Theory

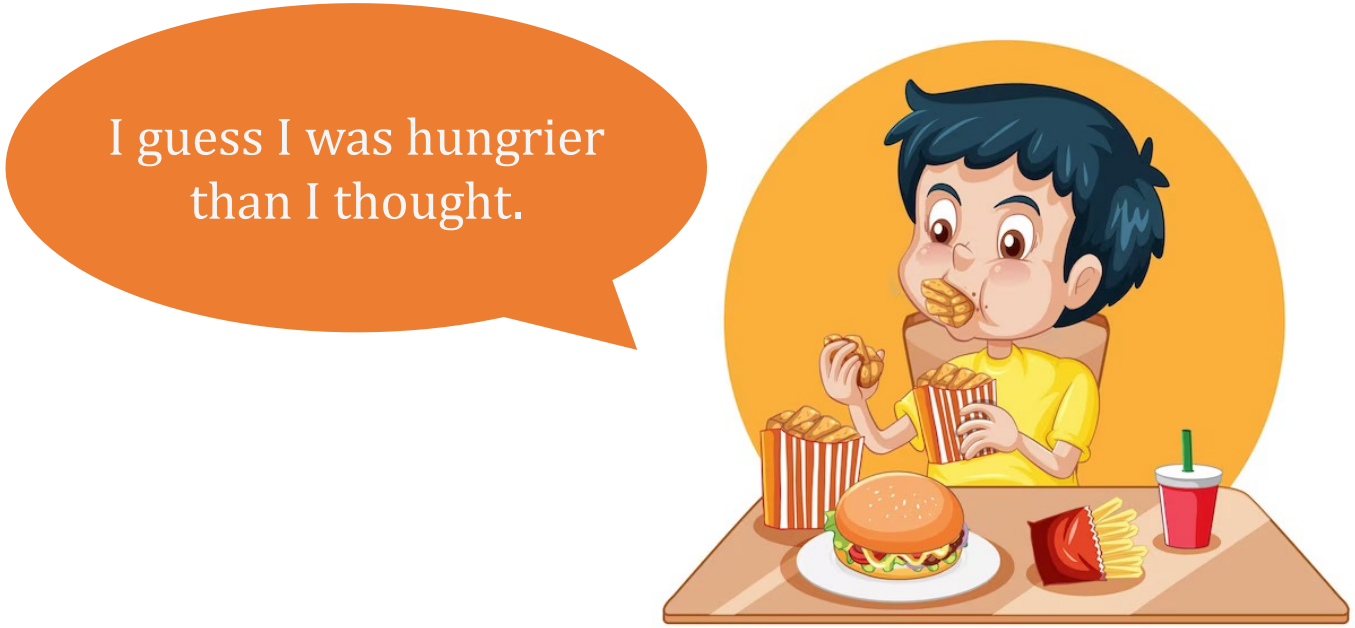
- People don't always come to know their own attitudes by introspecting about what they think or how they feel about something.
- Rather, they often look outward, at their behavior and the context in which it occurred, and **infer** what their attitudes must be.



- The theory that people come to know their own attitudes by looking at their behavior and the context in which it occurred and inferring what their attitudes must be.

Self-Perception Theory

- Just like we infer others' attitudes from their behaviors (social perception), we do the same with our own attitudes.
    - **Weak and ambiguous**



Explanation of Dissonance Effects

- **Self-Perception Theory** (Bem, 1967):
  - People aren't troubled by any unpleasant state of arousal like dissonance; they merely engage in a rational inference process.
  - They don't change their attitudes; rather, they infer what their attitudes must be.

- **Cognitive Dissonance Theory** (Fetinger, 1957):
  - Inconsistency between a person's thoughts, sentiments, and actions creates an aversive emotional state (dissonance) that leads to efforts to restore consistency.

Self-Perception Theory

- People value what they have chosen more after having chosen it because they infer that “If I chose this, I must like it.”
- People form tight bonds to groups that have unpleasant initiation rituals because they reason that “If I suffered to get this, I must have felt it was worth it.”



如果你看过我的任何关于
肯定或肢体语言

Goal

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    - Components of attitudes
    - How do we measure attitudes?
2) The attitude-behavior link
    - Attitudes to behavior
    - Behavior to attitudes
3) Cognitive dissonance theory
4) Self-perception theory



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